

KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY

Department of Energy Science and Engineering

B. Sc. Engineering 2nd Year 2nd Term Examination, 2019

ESE 2209

(Bio and Wind Energy Engineering)

Time: 3 Hours.

Full Marks: 210

N.B. i) Answer any THREE questions from each section in separate scripts.

ii) Figures in the right margin indicate full marks.

iii) Assume reasonable data if any missing.

SECTION – A

- 1(a). Based on the working principle classify anemometer. Discuss the working principle of cup anemometer and pressure tube anemometer. 3+8=11
- 1(b). Define average wind speed. On the basis of Weibull distribution function, prove that the average velocity of wind, $V_m = C\Gamma(1 + \frac{1}{k})$, where the symbols have their usual meaning. 2+10=12
- 1(c). Write the common methods for determining the Weibull parameters k and c . Also, briefly discuss all of methods. 3+9=12
- 2(a). With a schematic diagram, show the conversion of wind power into electricity. 04
- 2(b). Basis of the behaviour of rotor under varying operating conditions, discuss and explain the strip theory. 10
- 2(c). How is wind energy formed? Write the importance of using pitch controller in a wind turbine. 4+5=9
- 2(d). Why the output power of a wind turbine power plant is not stable without controller? What is FACTS devices and why it is used in wind turbine power plant? 6+6=12
- 3(a). What is life cycle analysis? Draw and discuss the various stages of life cycle of a wind turbine. 3+5=8
- 3(b). A land based wind turbine power plant consists of 15 wind turbine of 0.6 MW size. The cut-in, rated and cut-out velocities of the turbines are 3.5, 12.5 and 25.0 m/s, respectively. The Weibull shape and scale factors at the farm site are 2.8 and 8.2 m/s, respectively. The life period is 25 years, the capacity factor of the turbine is 0.335, the total energy requirement of the land based wind turbine is 3806.2 GJ). Calculate the energy payback ratio and energy payback period of the farm. 10
- 3(c). Explain the different ways in which the cost of a wind energy system is commonly expressed. 06
- 3(d). Discuss about the benefits of wind energy. Explain the factors which influence the wind energy economics. 5+6=11
- 4(a). What are the challenges in wind power generation? Define power coefficient and tip speed ratio. 4+3=7
- 4(b). With a neat sketch, explain the constructional features of a wind energy conversion systems considering various components. 12
- 4(c). With the help of schematic diagram, discuss the working principle of different types of variable speed wind generators. 12
- 4(d). What are the effects of change in supply frequency on torque and speed for wind turbine output. 04

SECTION – B

- 5(a). Explain the characteristics of a fuel. Compare the biofuel and fossil fuel. 4+3=7
- 5(b). Write the short note on different types of biofuel. 08
- 5(c). What are the potential raw materials for biomass briquetting? 05
- 5(d). With a schematic diagram, show the different ways of biomass briquetting. What are the factors that affect biomass briquetting? 8+7=15

- 6(a). What is bio-ethanol? What are the required conditions for ethanol fermentation? 2+5=7
- 6(b). Discuss in brief, the different growth phases of micro-organism for bio-ethanol production? 15
- 6(c). With a schematic diagram show the conversion of wheat into bioethanol. 07
- 6(d). Write short notes on: (i) E85G, (ii) E15D, (iii) B20 06
- 7(a). What is BOD? Is biochemical oxygen demand different from biological oxygen demand? 2+5=7
- 7(b). A secondary effluent has a total BOD₅ of 45 mg/L. A BOD₅ test is run on this same secondary effluent at the same time using a nitrification inhibitor. Using the data given below for this test, calculate the nitrogenous BOD. (Assume 300 mL BOD bottle is used to run this test). 06

Sample size	30 mL	50 mL	Blank
Initial DO (mg/L)	8.9	8.8	9.0
Final DO (mg/L)	6.0	4.2	8.9

- 7(c). What are the necessary conditions for anaerobic digestion of organic wastes? Discuss in brief. 10
- 7(d). Write short note on: fixed dome type biogas digester. 12
- 8(a). The C/N ratio of human excreta is about 8 and that of rice straw is 73. For optimum gas production, the required C/N ratio is 20-30:1. Therefore 1 kg human excreta, how much rice straw should be mixed? 07
- 8(b). What is gasification? Write short note on fluidized bed gasifier. 3+9=12
- 8(c). What is pyrolysis? What are the general mechanism of biomass pyrolysis? 2+8=10
- 8(d). Write short note on: Vacuum pyrolysis reactor. 06

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KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY

Department of Energy Science and Engineering

B. Sc. Engineering 2nd Year 2nd Term Examination, 2019

EE 2213

(Power Electronics)

Time: 3 Hours.

Full Marks: 210

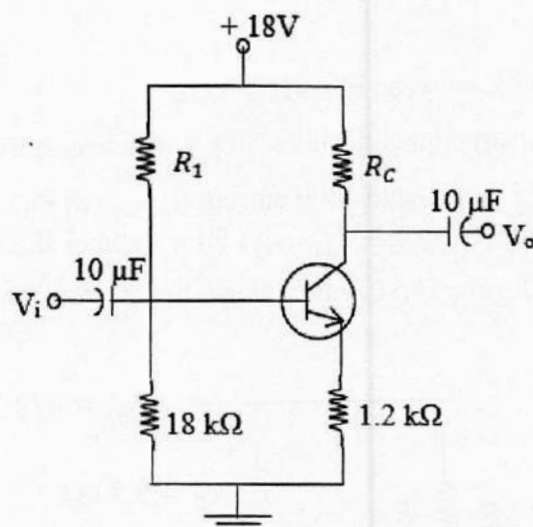
N.B. i) Answer any THREE questions from each section in separate scripts.

ii) Figures in the right margin indicate full marks.

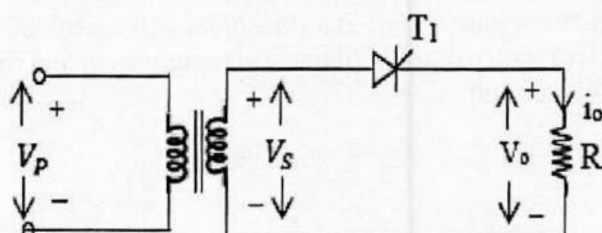
iii) Assume reasonable data if any missing.

SECTION – A

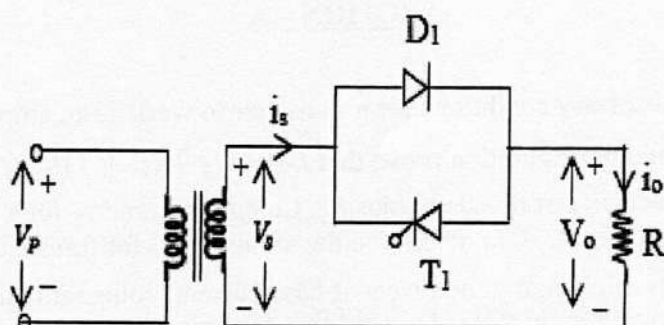
- 1(a). Write down the necessary conditions for a transistor to work as an amplifier or switch. 05
- 1(b). For common emitter configuration prove that $I_C = \beta I_B + (\beta + 1)I_{CB0}$. 10
- 1(c). Write down the necessity of transistor biasing. Calculate I_C and I_E for a transistor that has $\alpha_{dc} = 0.97$, $I_B = 120 \mu A$. Also determine the value of β_{dc} for transistor. 10
- 1(d). Explain the effects of changing the levels of base current, collector resistance and V_{CC} on the transistor operating point (Q) using load line analysis. 10
- 2(a). Given that $I_{CQ} = 2 mA$ and $V_{CEQ} = 10 V$, determine R_1 and R_C for the network of following figure. 10



- 2(b). What is meant by bias stabilization? Write down the effect of temperature on bias stabilization. 08
- 2(c). Derive the mathematical expressions of stability factor ($S(I_{CB0})$) for voltage divider bias configuration and hence show that voltage divider bias configuration is most stable. 17
- 3(a). What is power electronics? Why the study of power electronics is necessary for the students of ESE department? 10
- 3(b). Write the basic operation of single phase fullwave converter with circuit diagram and waveform. Also, compare various mode of operations. 15
- 3(c). A single phase halfwave converter has a purely resistive load of R and the delay angle is $\alpha = \frac{\pi}{2}$, determine the (i) rectification efficiency (ii) form factor (FF), (iii) ripple factor (RF) and (iv) peak inverse voltage (PIV) of thyristor T_1 . 10

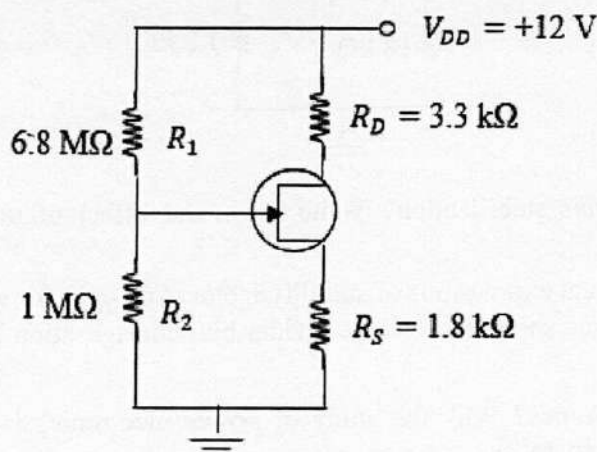


- 4(a). Draw the circuit diagram and necessary waveforms for speed control of a dc motor by a three phase semiconverter system for $\alpha = \frac{\pi}{2}$. 12
- 4(b). Draw the circuit diagram of three phase bidirectional controller with waveform of the output voltage. 11
- 4(c). A single phase ac voltage controller shown in figure has a resistive load of $R = 10 \Omega$ and the r.m.s input voltage is $V_s = 120 V$, $60 Hz$. The delay angle of thyristor T_1 is $= \frac{\pi}{2}$. Determine the (i) r.m.s value of output voltage V_o (ii) input power factor and (iii) Average input current. 12



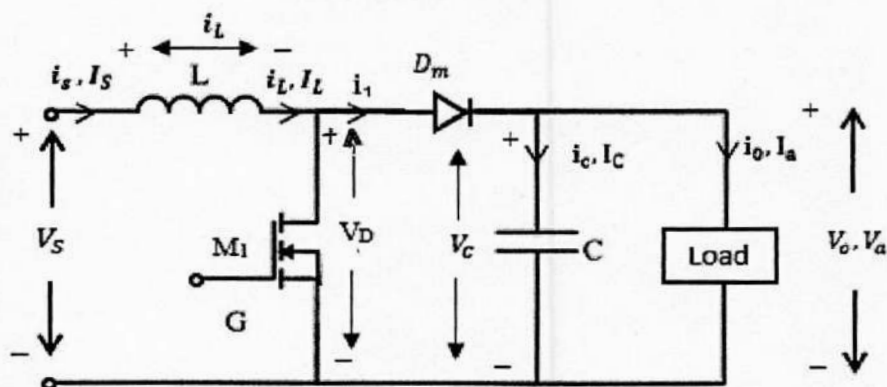
SECTION - B

- 5(a). What is FET? Distinguish between FETs and BJTs. 07
- 5(b). Briefly describe the constructional details of JFET with necessary diagrams. 10
- 5(c). Write short note on (i) Shorted-gate drain current (I_{DSS}), (ii) Pinch-off voltage (V_p), and (iii) Gate-source cutoff voltage ($V_{GS(off)}$) for n-channel JFETs. 09
- 5(d). Determine I_D and V_{GS} for the JFET with voltage divider in following figure ,given that $V_D = 7 V$. 09



- 6(a). Describe the structure of n-channel depletion type MOSFET. Also, explain the basic operation of it with its drain characteristics curve. 15
- 6(b). Show that greater the firing angle, the smaller is the average current for SCR half wave rectifier. 10
- 6(c). An SCR half-wave rectifier has a forward breakdown voltage of $150V$ when a gate current of $1 mA$ flows in the gate circuit. It a sinusoidal voltage of $400 V$ peak is applied, find: (i) firing angle, (ii) average output voltage, (iii) average current for a load resistance of 200Ω , and (iv) power output. 10

- 7(a). Explain why we need di/dt protection and $\frac{dv}{dt}$ protection and how to remedy these for an SCR. 10
- 7(b). What are the advantages and disadvantages of a buck-boost regulation? 06
- 7(c). Briefly explain the principle operation of step-up chopper and also show that by taking duty cycle $k = 0.5$, we can double the output voltage compared to input voltage. 09
- 7(d). A boost regulator shown in figure has an input voltage of $V_s = 5\text{ V}$. The average output voltage $V_o = 15\text{ V}$ and the average load current $I_a = 0.5\text{ A}$. The switching frequency is 25 kHz . If $L = 150\text{ }\mu\text{H}$ and $C = 220\text{ }\mu\text{F}$, determine the (i) duty cycle k , (ii) ripple current of inductor ΔI , and (iii) peak current of inductor I_2 . 10



- 8(a). Draw and explain the circuit diagram of three phase inverter formed by single phase inverters. 10
- 8(b). Describe the principle operation of single phase bridge inverter with it's related circuit diagram and waveforms. 10
- 8(c). Draw a 3 phase inverter with six transistor and six diodes. Also draw its waveform for (i) phase voltages, and (ii) line voltages for 120 degrees conduction. 10
- 8(d). Why voltage control is necessary for inverters? 05

KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY

Department of Energy Science and Engineering

B. Sc. Engineering 2nd Year 2nd Term Examination, 2019

Hum 2213

(Economics and Accounting)

Time: 3 Hours.

Full Marks: 210

N.B. i) Answer any THREE questions from each section in separate scripts.

ii) Figures in the right margin indicate full marks.

iii) Assume reasonable data if any missing.

SECTION – A

- 1(a). Define Economics. Distinguish between Microeconomics and Macroeconomics. 15
- 1(b). What is demand? What determines the quantity of a good that buyers demand? Explain. 20
- 2(a). What is indifference curve? Discuss the characteristics of indifference curve. 15
- 2(b). Pharmaceutical drugs have an inelastic demand and computers have an elastic demand. Suppose that technology advancement doubles the supply of both products (that is, the quantity supplied at each price is twice what it was). Explain, what happens to the equilibrium price and quantity in each market? 20
- 3(a). What is production function? Explain the relationship between total product and marginal product. 15
- 3(b). What is meant by a “Competitive firm”? Under what conditions will a firm exit a market? Explain. 20
- 4(a). What is gross domestic product (GDP), real GDP, and GDP growth. 15
- 4(b). What is inflation? What are the main causes of inflation in a country like Bangladesh? 10
- 4(c). What is private savings, public savings and national savings? Explain. 10

SECTION – B

- 5(a). Define accounting. Why accounting is called the language of business? 10
- 5(b). Discuss the principle of accounting. 15
- 5(c). Briefly describe the double entry system of accounting. 10
- 6(a). Define depreciation. What are the factors considered in computing depreciation? 15
- 6(b). What is transaction? Write down the characteristics of a transaction. 10
- 6(c). What is adjusting entries? Discuss the objectives of adjustments. 10
- 7(a). Distinguish between journal and ledger. 10
- 7(b). Some transactions of Samir Company during the month of April are given below: 25
- April 1: Samir invested cash, Tk. 50,000 and equipment of Tk. 50,000 in his business.
- April 2: Appointed an assistant whose salary will be Tk. 5,000.
- April 5: Paid Tk. 1,000 for two months' rent in advance for an office.
- April 10: Purchased furniture for Tk. 10,000 paying 50% in cash and the balance on account.
- April 12: Billed customer for service performed Tk. 5,000.
- April 15: Paid Tk. 2,000 cash on balance owed for furniture.
- April 20: Received Tk. 2,000 from customer previously billed.
- April 25: Samir withdrew Tk. 1,200 cash for personal use.
- April 28: Paid utilities bill Tk. 500.
- April 30: Transfer one month's rent to rent expense account.
- Requirement: Record the above entries in general journal.

8. The unadjusted trial balance of Arnold Company as of December 31, 2018 is given below:

Arnold Company
Trial Balance
December 31, 2018

Name of Accounts	Debit (Tk.)	Credit (Tk.)
Cash	15,500	
Prepaid insurance	1,400	
Supplies	24,000	
Equipment	28,000	
Accumulated depreciation equipment		6,500
Capital		37,000
Withdrawals	5,300	
Service revenue		82,900
Salary expense	42,000	
Rent expense	10,200	
Total	1,26,400	1,26,400

- (i) Expired insurance TK. 800.
- (ii) Unused supplies inventory Tk. 5,100.
- (iii) Estimated depreciation of equipment Tk. 1,300.
- (iv) Salaries earned but not paid Tk. 500.

Required:

- (a) Statement of comprehensive income for the year ended December 31, 2018. 20
- (b) Statement of financial position as on December 31, 2018. 15

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KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY

Department of Energy Science and Engineering

B. Sc. Engineering 2nd Year 2nd Term Examination, 2019

Math 2213

(Complex Variables and Fourier Analysis)

Time: 3 Hours.

Full Marks: 210

N.B. i) Answer any THREE questions from each section in separate scripts.

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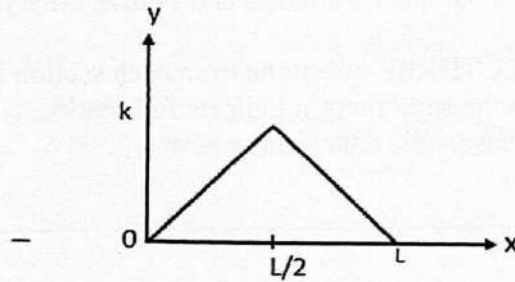
iii) Assume reasonable data if any missing.

SECTION – A

- 1(a). If $z = \frac{z_1 z_2}{z_3}$, where $z_1 = 3 + 4i$, $z_2 = 2 - i$ and $z_3 = 2 + 3i$, find $|z|$ and $\arg(z)$. 08
- 1(b). Solve $z^6 + 1 = \sqrt{3}i$ and locate them graphically in the finite z plane. 09
- 1(c). If $w = f(z)$ is continuous at $z = z_0$, then show that $\bar{w}$ is also continuous at $z = z_0$. 09
- 1(d). If $|z_1 + z_2| = |z_1 - z_2|$, then prove that the difference of the amplitude of z_1 and z_2 is $\frac{\pi}{2}$. 09
- 2(a). Define analytic function. Show that the function $f(z) = \frac{1}{z}$ is analytic everywhere except at the origin. 10
- 2(b). Find the constants a, b, c such that the function $f(z) = x^2 - xy - y^2 + i(ax^2 + 2bxy + cy^2)$ is analytic. Also express $f(z)$ in terms of z . 12
- 2(c). Show that the function $f(z) = u + iv$, where $f(z) = \begin{cases} \frac{x^3(1+i) - y^3(1-i)}{x^2 + y^2}, & z \neq 0 \\ 0, & z = 0 \end{cases}$ satisfies the Cauchy-Riemann conditions at $z = 0$. Is this function analytic at $z = 0$? Justify your answer. 13
- 3(a). Evaluate $\oint_C \frac{3z-1}{z^3+z} dz$, where C is a square $ABCD$ having vertices at $A(2,0)$, $B(0,2)$, $C(-2,0)$ and $D(0,-2)$ respectively in the positive sense. 10
- 3(b). Locate and name the singularities of $f(z) = \sec z$. 05
- 3(c). What is meant by harmonic function? Prove that $u = \ln(x^2 + y^2)$ is harmonic function. 10
- 3(d). State the Cauchy's integral formula. Apply this theorem to evaluate $\oint_C \frac{\sin \pi z^2 + \cos \pi z^2}{(z-1)(z-2)} dz$, where C is circle $|z| = 3$. 10
- 4(a). Show that Bessel's function of various order can be obtained from the coefficient of different powers of t in the expansion of $e^{\frac{x}{2}(t - \frac{1}{t})}$. 16
- 4(b). Using generating function of Bessel's function show that $\sin(x \sin \theta) = 2[\sin \theta \cdot J_1(x) + \sin 3\theta \cdot J_3(x) + \sin 5\theta \cdot J_5(x) + \dots]$. 09
- 4(c). Express $J'_2(x)$ in terms of $J_0(x)$ and $J_1(x)$, where $J_n(x)$ is the Bessel's function of first kind. 10

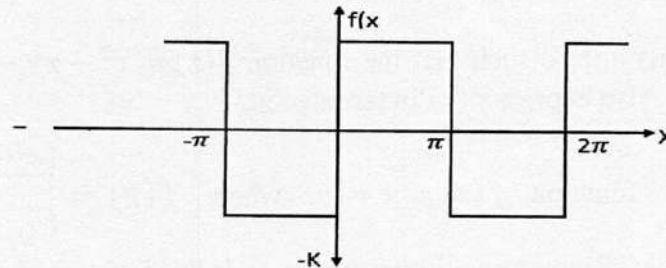
SECTION – B

- 5(a). What is meant by the Fourier series? Find the two half- range Fourier expansion of the function graphed in figure. 15



- 5(b). Find the Fourier trigonometric series of the periodic function $f(x) = \begin{cases} 0, & \text{if } -\pi < x < 0 \\ x, & \text{if } 0 < x < \pi \end{cases}$ and $f(x + 2\pi) = f(x)$. 15
- 5(c). Write down the Euler's formula for representing a function in series in the interval $\alpha < x < \alpha + 2\pi$. State the assumptions for the expansion of a function $f(x)$ in Fourier series. 05
- 6(a). Find the complex form of the fourier series of the periodic function $f(x)$ of period 2. which is defined by $f(x) = e^{-x}$, $-1 < x < 1$. 15
- 6(b). Determine the Fourier coefficients of periodic function $f(x)$ which is graphed in figure and mention the name of the wave. Also write the Fourier series of $f(x)$ and hence show that 20

$$1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \dots = \frac{\pi}{4}.$$



- 7(a). Classify the following partial differential equation : 06
- (i) $\frac{\partial^2 u}{\partial x^2} = 5 \frac{\partial u}{\partial x} + \frac{\partial u}{\partial y}$
- (ii) $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial x \partial y} + \frac{\partial^2 u}{\partial y^2} = 0$
- (iii) $\frac{\partial^2 u}{\partial x \partial y} = 3 \frac{\partial u}{\partial y}$
- 7(b). The temperature distribution in a bar of length π , which is perfectly insulated at the ends $x = 0$ and $x = \pi$ is governed by the partial differential equation $\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}$. Assuming the initial temperature distribution as $u(x, 0) = \cos 3x$, find the temperature distribution at any instant of time. 14
- 7(c). Solve the heat equation $\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}$ under the condition: 15
- $u(x, 0) = 3 \sin n\pi x$ for $0 < x < 1$
- $u(0, t) = u(1, t) = 0$ for $t > 0$

- 8(a). Determine the solution of two dimensional Laplace's equation in cylindrical coordinates. 15
- 8(b). For a semi-circular plate of radius 'a' with boundary diameter at 0°C and surface at 100°C , show that the temperature distribution is given by 15

$$u(0, t) = \frac{400}{\pi} \sum_{n=1}^{\infty} \frac{\pi^{2n-1}}{2n-1} \cdot \frac{\sin(2n-1)\theta}{a^{2n-1}}.$$

- 8(c). Verify that $T(x, t) = e^{-kt} \sin x$, $k > 0$ is a solution of the parabolic heat equation $\frac{\partial T}{\partial t} = k \frac{\partial^2 T}{\partial x^2}$. 05

KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY

Department of Energy Science and Engineering

B. Sc. Engineering 2nd Year 2nd Term Examination, 2019

ME 2213

(Dynamics and Kinematics of Machineries)

Time: 3 Hours.

Full Marks: 210

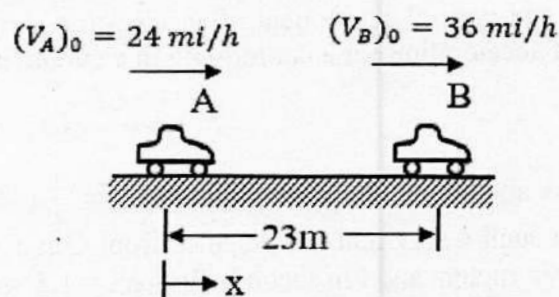
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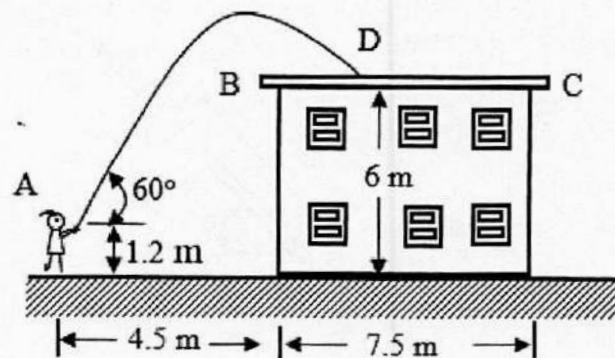
iii) Assume reasonable data if any missing.

SECTION – A

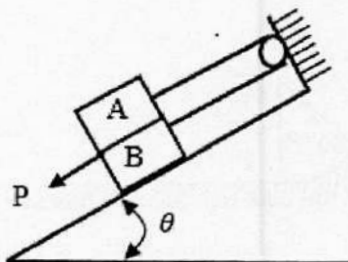
- 1(a). How dynamics is related to energy science? Explain. 05
- 1(b). Automobiles A and B are travelling in adjacent highly lanes and at $t = 0$ have the positions shown along with the speeds. Knowing that automobile A has a constant acceleration of 0.55 m/s^2 and that B has a constant deceleration of 0.37 m/s^2 . Determine – 14
- (i) when and where A will overtake B.
- (ii) the speed of each automobile at that time.



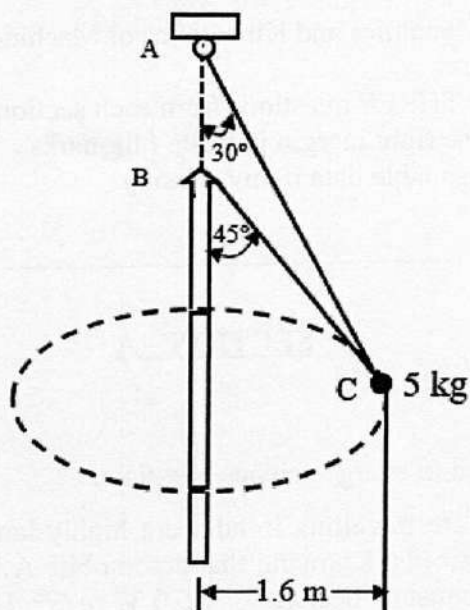
- 1(c). A nozzle at A discharges water with an initial velocity of 12 m/s at an angle of 60° with the horizontal. Determine where the stream of water strikes the roof. Check that the stream will clear the edge of the roof. 16



- 2(a). Block A has a mass of 40 kg and block B has a mass of 20 kg. The coefficient of friction between all surfaces of contact are $\mu_s = 0.15$ and $\mu_k = 0.10$. Knowing that $\theta = 30^\circ$ and the magnitude of force P applied to block B is 280 N, determine – 17
- (i) the acceleration of block A.
- (ii) the tension in chord.

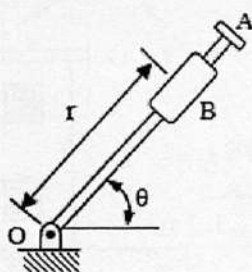


- 2(b). A single wire ACB passes through a ring at C attached to a sphere which revolves at a constant speed of V in the horizontal circle shown. Knowing that the tension is the same in both portions of the wire, determine the speed, V . 11

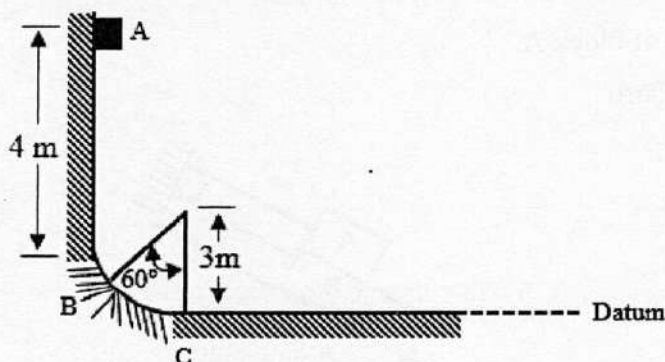


- 2(c). What does tangential and normal component of acceleration signify? Draw the tangential and normal components of acceleration for a car moving in a curved path. 07

- 3(a). The rotation of rod OA about O is defined by relation $\theta = \frac{1}{2} (4t - 3t^2 - 1)$ and the collar B slides along the rod in such a way that its distance from O is $r = 1.25t^2 - 0.9t^3$, where θ is expressed in radians, r in meters and t in seconds. When $t = 1.5$ second, determine – 17
- (i) the velocity of collar.
 - (ii) the total acceleration of the collar.
 - (iii) the acceleration of the collar relative to the rod.

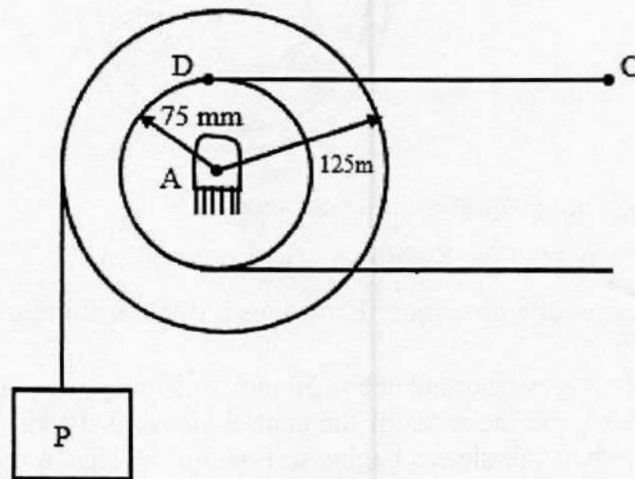


- 3(b). A 1.25 N, pellet is released from rest at A and slides without friction along the surface shown. Determine the force exerted by the surface on the pellet as it passes through (i) point B, and (ii) point C. 18

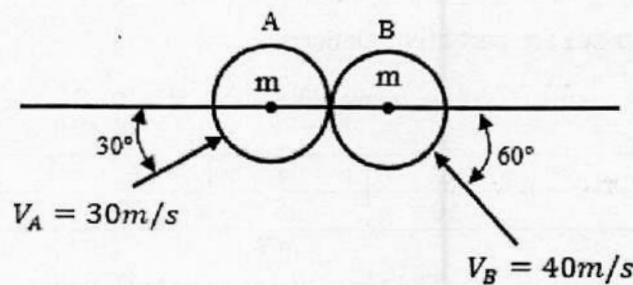


- 4(a). The load P is connected to a double pulley by one of the two inextensible cables as shown. The motion of the pulley is controlled by cable C, which has a constant acceleration of 225 mm/s^2 and an initial velocity of 300 mm/s both directed to right. Acceleration of D is also 225 mm/s^2 . Determine — 19

- the number of revolutions executed by the pulley in 2 sec.
- the velocity and change in position of the load B after 2 sec.
- the acceleration of point D on the rim of inner pulley at $t = 0$ sec.



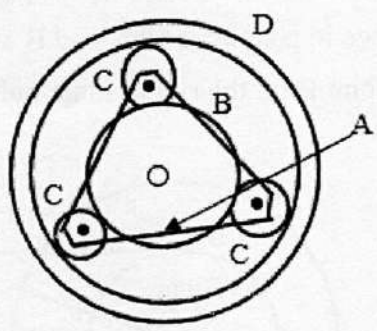
- 4(b). The magnitude and direction of the velocities of two identical frictionless balls before they strike each other are as shown. Assuming $e = 0.90$, determine the magnitude and direction of the velocity of each ball after the impact. 16



SECTION – B

- Prove that the maximum fluctuation of energy in a flywheel can be expressed as, $\Delta E = E \times 2C_s$, where E is the mean kinetic energy of the flywheel and C_s is the coefficient of fluctuation speed. 10
- Discuss the effect of partial balancing of reciprocating parts of two cylinder locomotives. 12
- An Otto cycle engine develops 50 kW at 150 rpm with 75 explosions per minute. The change of speed from the commencement to the end of power stroke must not exceed 0.5% of mean on either side. Find the mean diameter of the flywheel and a suitable rim cross section having width four times the depth so that the hoop stress does not exceed 4 MPa. Assume that flywheel stores 16/15 times the energy stored by the rim and the work done during power stroke is 1.40 times the work done during the cycle. Density of rim material is 7200 kg/m^3 . 13
- What is the significance of tuning moment diagram of an IC engine? 05
- Show that "The common normal at the point of contact between a pair of teeth must always pass through the pitch point". 13

6(c). In an epicyclic gear train of the “sun and planet type” as shown in figure the pitch circle diameter of internally toothed ring D is to be 216 mm and the module 4 mm. When the ring D is stationary the spider A, which carries three planet wheels C of equal size, is to make one revolution in the same sense as the sun wheel B for every five revolution of the driving spindle carrying the sun wheel B. Determine suitable number of teeth for all the wheels and the exact diameter of pitch circle of the ring.



- 7(a). Explain the following properties of a governor; 09
 (i) Sensitiveness (ii) Stability (ii) Isochronism.
- 7(b). What is the function of a governor? How does it differ from that of a flywheel? 08
- 7(c). The arms of a porter governor are each 250 mm long and pivoted on the governor axis. The mass of each ball is 5 kg and the mass of the central sleeve is 30 kg. The radius of rotation of each ball is 150 mm when the sleeve begins to rise and reaches a value of 200 mm for maximum speed. Determine the speed range of the governor. If the friction at the sleeve is equivalent of 20 N of load at the sleeve, determine how the speed range is modified. 18
- 8(a). Justify the statement “If the resultant of all forces is zero, then there will be no unbalanced force, but even a shaking couple would be present”. 08
- 8(b). Why reciprocating masses are partially balanced? 10
- 8(c). A compound gear train consists of six gears. The number of teeth are as follows. 17

A	B	C	D	E	F
60	40	50	25	30	24

The gear B and C are on one shaft. The Gear A drives gear B, gear C drives gear D and gear E drives gear F. If the gear A transmits 1.5 kW at 100 rpm and the gear train has an efficiency of 80 percent, find the torque on gear F.

— END —